Individual Work Breakdown Report

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Project Context

This document summarizes my individual contribution to the group project "Dublin Housing Affordability Analysis (2015–2025)." The project objective was to build a complete ETL and visualization pipeline using multiple datasets, dual database systems, and reproducible analytics methods.

Primary Ownership Areas

- Transformation pipeline design and implementation.
- Data cleaning, validation, and harmonization.
- Metric engineering for affordability and growth analysis.
- ETL orchestration review and processed output verification.

Detailed Contributions Across Project Stages

1. Data Understanding and Preparation Planning

I participated in initial source review and variable mapping, focusing on how raw property and API datasets could be integrated into a coherent yearly affordability model. I helped define field expectations, transformation sequence, and failure-handling priorities.

2. Transformation Pipeline Development

I contributed significantly to `transform.py` logic, including:

- date and numeric standardization,
- county and analysis-window filtering,
- schema normalization across variable source labels,
- annual aggregation for price and income metrics,
- affordability index and growth metric calculations.

This stage was central to converting noisy, heterogeneous source data into analysis-ready structures.

3. Robustness Enhancements

I helped address multiple data quality challenges:

- fallback parsing for inconsistent JSON-stat region/time labels,
- controlled behavior when expected labels are missing,
- area extraction improvements to avoid dominant unknown categories,
- low-volume area thresholding to reduce unstable estimates.

These changes improved reliability and interpretability of downstream results.

4. Curated Data and Database Integration

I supported integration between transformation outputs and relational storage. I validated schema compatibility, table loads, and deterministic overwrite behavior. During debugging, I participated in resolving a string-length truncation issue by adjusting area field width in PostgreSQL schema logic.

5. Analytical Output Validation

I checked consistency of processed outputs across reruns, including:

- year range validity,
- affordability overlap behavior,
- area summary distributions,
- insight table coherence with computed indicators.

I also reviewed whether chart inputs accurately reflected transformed metrics.

6. Reporting and Presentation

I co-authored methodology and evaluation narratives, especially sections explaining transformation decisions and why those choices were necessary under real-world data constraints. I contributed to presentation script segments describing ETL logic and processing rationale.

Cross-Stage Evidence of Participation

- **Selection:** contributed to variable relevance review.
- **Collection:** validated extracted structures for transform readiness.
- **Storage:** reviewed curated schema fit and load behavior.
- **Processing:** primary owner of cleaning and feature engineering logic.
- **Analysis:** verified interpretation consistency with transformed outputs.
- **Communication:** co-developed methods and evaluation write-up.

Challenges and Responses

Challenge 1: Inconsistent labels and dimensions in API-derived tables. **Response:** Implemented fallback normalization and parsing resilience to avoid brittle transformations.

Challenge 2: Loss of area granularity due to sparse Eircodes. **Response:** Added address-based fallback extraction and group threshold filtering.

Challenge 3: Curated schema mismatch after transformation enhancement. **Response:** Coordinated schema update and rerun validation to restore successful end-to-end loading.

Reflection

My main contribution was ensuring that transformed data remained both analytically meaningful and operationally stable. By focusing on cleaning logic, robust parsing, and metric engineering, I helped convert fragmented source records into trustworthy affordability outputs suitable for reporting and dashboarding.

Declaration

This report is an original account of my own contribution to the team project.